

Cheah Hao Yi

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SKILLS

Programming: Python, C/C++, Bash, Java, Javascript

Cybersecurity: CIS Benchmarks, VAPT, MITRE Att&ck framework, Threat Modeling (STRIDE), CVE Remediation

Network: TCP/IP, OSPF, ISIS, BGP, SSL/TLS, Firewall/Router/Switch configuration, IPS

AI Agents: Local LLM Deployment, Retrieval-Augmented Generation (RAG), Loop Engineering

PROFESSIONAL EXPERIENCE

Network Security Engineer | Huawei International Pte. Ltd 2025 - 2026

- Secured Singapore Critical Information Infrastructure (CII) serving 1.2M users by driving end-to-end CVE vulnerability patching across enterprise servers, storage arrays, and network appliances achieving 100% SLA compliance.
- Hardened EulerOS on virtualized IP Multimedia System (vIMS) against potential attack vectors by ensuring compliance of over 200 CIS benchmark best practices across over 50 virtual hosts.
- Automated security reporting and workflow pipelines by scripting over 50 Python and Bash tools, reducing manual administrative overhead by 80%, improving team efficiency and project execution.
- Enforced operational risk controls and zero-downtime execution by authoring Methods of Procedure (MOP) and Requests for Change (RFC) for mission-critical network deployments, acquiring endorsement from management.
- Bridged cross-functional technical and security requirements between customer and internal R&D teams in English and Chinese, accelerating deployment cycles and driving customer satisfaction outcomes.

PROJECTS

AISPloitable | Autonomous Threat Intelligence and Empirical Sandbox Verification Platform

- Engineered an autonomous vulnerability triage agent in Python using local Gemma 4 open-weight models (gemma4:e2b/e4b) to automate CVE analysis, multi-step threat modeling, and severity assessment.
- Orchestrated a deterministic dual-container Docker sandbox on an isolated bridge subnet within a local environment, enabling safe execution of dynamic PoC scripts with zero localhost loopback false positives.
- Integrated an offline threat intelligence RAG engine indexing 755 MITRE ATT&CK and MITRE ATLAS techniques using in-memory token overlap and Jaccard similarity scoring without external DB dependencies.
- Built a real-time telemetry streaming dashboard displaying synchronized attacker-victim container output, attack graphs, and automated executive Markdown/PDF audit report exports, providing relevant remediation recommendations.

Proxmox Multi-Tenant Security HomeLab | Local-LLM Orchestration

- Built a resilient, fault-tolerant LLM pipeline in Python/Bash on a headless Ubuntu Server, featuring graceful SIGTERM handling, state persistence, and automatic crash recovery on resource constrained CPU-only systems.
- Mitigated hypervisor kernel panics in legacy target VMs (Kernel 3.13 / Metasploitable 3) for safe VM deployment.
- Architected an isolated offensive security lab (Kali Linux, vulnerable Linux/Windows targets) to practice memory corruption, privilege escalation, and exploit execution in preparation for the OSCP certification.

TEACHING EXPERIENCE

National University of Singapore (NUS) | Computer Science Teaching Assistant

- Instructed over 50 undergraduate and Master's students on Data Structures & Algorithms (CS2040C, IT5003) in Python and C++, teaching Big-O analysis, graph theory, heaps, hash tables, and dynamic programming.
- Mentored over 30 students on Programming Methodology (CS1010) in C, leading hands-on labs on low-level memory management, pointer arithmetic, Unix CLI toolchains, and Vim/Git workflows.
- Guided multiple collaborative software engineering teams (CS2113) in Java, conducting code reviews and teaching object-oriented design patterns, software architecture, system design and Git-based SDLC practices.

EDUCATION, CREDENTIALS

National University of Singapore (NUS) | B.Eng in Computer Engineering with Honours (Distinction) 2021 - 2025

- Full ride ASEAN Undergraduate Scholarship, Second Major in Business Management (Finance), Specialization in IoT
- CISSP (Associate of ISC2), Cisco Certified Network Associate (CCNA), Red Hat Certified System Administrator (RHCSA), Juniper Networks Certified Associate (JNCIA-Junos), Certified Network Cabling Design Professional (CNCDP), Huawei Certified IT Associate - Storage, Datacom (HCIA)